

ML Observability Pipeline

Project Structure Overview

Folder layout, contents, and purpose — as of 2026-06-04, post-Item-6 production regen. Cross-checked against context/_global.md, context/_interfaces.md, context/model.md, context/drift.md, DEV_NORMS.md, and ADRs 0001/0005/0008/0009.

Repository tree

ML Observability Pipeline/

│

├── PLAN.md.docx

├── HANDOFF.md

├── ACCOUNT_SETUP.md.docx

├── DEV_NORMS.md

├── requirements.txt

├── docker-compose.yml

│

├── context/

├── docs/

│ ├── adr/

│ ├── sessions/

│ ├── next_session_brief.md

│ ├── image_prompts_progress_2026-05-29.md

│ └── podcast_series_prompt.md

│

├── review_packets/

├── review_responses/

├── templates/

├── scripts/

│

├── shared/

│ ├── features.py

│ ├── score.py

│ └── drift.py

│

├── simulator/

│ ├── pump.py

│ ├── scenario.py

│ ├── publisher.py

│ ├── runner.py

│ ├── config.{yaml,aws-iot.yaml,example.yaml}

│ ├── .secrets/P-00/

│ └── tests/

│

├── model/

│ ├── train.py

│ ├── artifacts/

│ │ ├── model.pkl

│ │ ├── operational_reference_distribution.json (4-feature PSI, ~2.4 KB)

│ │ └── README.md

│ └── tests/

│

├── local_runtime/

│ ├── service.py

│ ├── subscriber.py

│ ├── window.py

│ ├── influx_writer.py

│ ├── config.{yaml,example.yaml}

│ └── tests/

│

├── lambda_scorer/

├── lambda_s3_batcher/

├── infra/

└── dashboards/

Master project plan (referenced as PLAN.md §X)

Project overview (§1-10): scenario, AWS rules, decisions, o

AWS account setup notes (one-time)

Roles, workflow, context-loading tiers, doc conventions

Python deps

Local infra: Mosquitto + InfluxDB + Grafana

Per-component context docs (Tier 2/2b modular load)

Documentation root

9 ADRs, immutable once accepted

12 session logs (2026-05-24 thru 2026-06-04)

Rolling session-to-session handoff

Outgoing: questions for Gemini reviewer

Incoming: Gemini's responses

Scaffolds (ADR, session log, packet, brief)

gemini_review.{ps1,sh} (direct-API per ADR 0001)

*** PARITY BOUNDARY *** (ADR 0005)

FEATURE_NAMES (8) + PSI_FEATURE_NAMES (4)

score(features) -> float

load_reference + compute_psi (ADR 0007)

Synthetic pump fleet

Physics + state machine (ADR 0002)

Fault injection (ADR 0004)

aiomqtt client (ADR 0003)

asyncio orchestrator

config.{yaml,aws-iot.yaml,example.yaml}

AWS IoT per-Thing cert bundle (one Thing per pump)

10 test files

Training + shipped artifacts

python -m model.train --n-pumps 30 --seed 0

Shipped binaries (PO commits these)

joblib bundle, 8-feature scorer (~300 KB native)

(4-feature PSI, ~2.4 KB)

Sandbox-vs-production build distinction

test_train + test_score_wiring

Local-mode scorer service

MQTT sub -> feature window -> score+PSI -> Influx

aiomqtt subscriber, one connection per pump

Per-pump rolling 1-hour deque

13-field points on compute ticks (ADR 0009)

Structural-parity guards live here

[planned -- context-only, code not built]

[planned -- context-only, code not built]

[planned -- context-only, code not built]

[planned -- context-only, code not built]

Project convention call-out

scripts/gemini_review.ps1 + .sh are the canonical review tools per ADR 0001 (Direct Gemini API for reviews). They deliberately bypass the gemini CLI — the CLI's positional-vs-flag confusion and 503 backoff are exactly why the project standardized on direct HTTPS POSTs to the generateContent endpoint. For Step 2 of any session's review loop:
`.\scripts\gemini_review.ps1 -Slug <slug>.`

Folder purposes

Root files

PLAN.md.docx and ACCOUNT_SETUP.md.docx are Word docs (candidates for the same docx-to-md cleanup HANDOFF.md just went through). DEV_NORMS.md is the collaboration playbook — 9 sections covering Roles, Project north stars, Workflow, Gemini review loop, Modular context (Tier 1/2/2b), Documentation method, Git conventions, Definition of session done, and Norm conflicts. No top-level README.md or top-level tests/ — tests live with their components by convention.

context/

One markdown file per component plus cross-cutting. **_global.md** (north stars + tech locks + cross-component invariants) and **_interfaces.md** (cross-component data shapes) are the Tier 1 always-loaded set. **simulator.md**, **model.md**, **drift.md**, **local_runtime.md**, **lambda_scorer.md**, **lambda_s3_batcher.md**, **infra.md**, **dashboards.md** are Tier 2/2b component docs — loaded per session. **README.md** documents the tiering rules. Pattern: context doc precedes code (the four planned modules have context docs but no code yet).

docs/adr/

Architecture Decision Records, 9 of them. Immutable once accepted; superseded ADRs get amendments, not edits. Numbering: 0001 direct-Gemini-API, 0002 RPM coupling, 0003 aiomqtt per-pump, 0004 tick-driven scenarios, 0005 shared package + parity boundary, 0006 model family + features, 0007 PSI implementation + cadence, 0008 operational reference source-separated, 0009 PSI surface != scorer feature set.

docs/sessions/

Twelve session logs (2026-05-24 through 2026-06-04). One per work session. Each captures: intent, decisions, what changed (code/artifacts/tests/docs), Gemini review highlights, open follow-ups, and a note-for-next-session. Per DEV_NORMS §6.2.

docs/{next_session_brief, image_prompts_progress_..., podcast_series_prompt}.md

Three standalone docs in docs/. **next_session_brief.md** is the rolling session-to-session handoff — always reflects what the NEXT session should pick up. **image_prompts_progress_2026-05-29.md** tracks prior AI image-gen prompts for the architecture diagram. **podcast_series_prompt.md** is a creative side-thread (project-narrative podcast prompts).

review_packets/ + review_responses/

Paired files per session, 1:1 by date+slug. Packet = what we send Gemini (Summary + Diff + Specific questions + What I'm NOT looking for + Resolution table). Response = Gemini's answer. The Resolution table in the packet gets filled in by Claude after Gemini responds. Per DEV_NORMS §6.3.

templates/

Four scaffolds: ADR template, session log template, review packet template, session brief template. New session logs/packets get copied from these as the starting point.

scripts/

gemini_review.ps1 (PowerShell) and gemini_review.sh (bash) — direct-API wrappers per ADR 0001. Reads a review packet, POSTs to Gemini's generateContent endpoint, writes the response. Fallback model flag: -Model gemini-flash-latest if -Model gemini-pro-latest hits 503s.

shared/ — THE PARITY BOUNDARY (ADR 0005)

Imported identically by local_runtime (today) and the future lambda_scorer (planned). No vendoring; both modes import as peers. **features.py** exports FEATURE_NAMES (8 — scorer input) and PSI_FEATURE_NAMES (4 — drift surface, strict subset per ADR 0009); extract_features produces the 8-feature dict. **score.py** wraps joblib-loaded model behind score(features) -> float. **drift.py** exposes load_reference + compute_psi (ADR 0007/0008/0009).

simulator/

Tick-driven pump fleet. Publishes telemetry to MQTT (local mode, Mosquitto in Docker) or AWS IoT Core (AWS demo mode). **pump.py** implements the HEALTHY/DEGRADING/FAILING/FAILED state machine + the RPM-coupled-to-degradation physics (ADR 0002). **scenario.py** runs fault injection (ADR 0004). **publisher.py** is the

aiomqtt-based MQTT client (ADR 0003). **runner.py** orchestrates N pumps under asyncio. Three config files: yaml for local, aws-iot.yaml for AWS demo, example.yaml as a template. **.secrets/P-00/** holds the per-Thing AWS IoT certificate bundle (one Thing per pump). **tests/** has 10 test files.

model/

Training + shipped artifacts. **train.py** generates a 30-pump training corpus, fits a HistGradientBoostingClassifier (ADR 0006), and emits both the model and the operational PSI reference. Default flag `--reference-source operational` per ADR 0008. **artifacts/** committed: model.pkl (joblib, FEATURE_NAMES = 8), operational_reference_distribution.json (PSI_FEATURE_NAMES = 4 per ADR 0009), README.md (sandbox-vs-production build distinction, new 2026-06-04). **tests/** has test_train.py + test_score_wiring.py.

local_runtime/

Local-mode scorer service. **service.py** wires the MQTT subscriber to the per-pump feature window to score + PSI to the InfluxDB writer. **subscriber.py** is the aiomqtt subscriber (one connection per pump). **window.py** maintains the per-pump rolling 1-hour feature deque. **influx_writer.py** emits InfluxDB points (13 fields per compute tick post-ADR-0009; 9 fields on non-compute ticks). **tests/** hosts the structural-parity guards: test_structural_parity_no_vendoring (no copies of shared/), test_structural_parity_score_loads_from_shared, test_structural_parity_compute_psi_loads_from_shared, and test_psi_feature_names_is_subset_of_feature_names (ADR 0009 asymmetry pin).

lambda_scorer/, lambda_s3_batcher/, infra/, dashboards/ [planned, not built]

Context docs exist (context/{lambda_scorer,lambda_s3_batcher,infra,dashboards}.md) but the CODE directories are not yet created. Project pattern: context doc precedes code. Next session up is most likely lambda_scorer — per the brief, every dependency is now in production shape.

Tech stack at a glance

From `context/_global.md §Tech locks`. These are pinned choices — changes require an ADR.

Concern	Pick	Rationale (1 line)
Language	Python 3.12	Pip + venv; sklearn ecosystem
IaC	Terraform	Schema-in-code; no Glue Crawler
Hot store	DynamoDB	Timestream closed to new accts 2025-06
Archive	S3 + Glue Catalog	Schema-in-Terraform; no Crawler cost
Batching	Lambda + EventBridge	Firehose has no Always-Free tier
Local TSDB	InfluxDB OSS (Docker)	Free; same query semantics as Cloud
Dashboards	Grafana local (Docker)	Managed Grafana costs \$9/user/month
Broker (local)	Mosquitto (Docker)	MQTT v3.1.1; no auth in local mode
Drift metric	PSI	PLAN.md §2.7; Laplace alpha=1.0 (ADR 0007)
Model	HistGradientBoostingClassifier	sklearn; AUC 0.998 on synthetic (ADR 0006)
Region	eu-central-1 ONLY	Cross-region transfer costs add up

Anti-patterns (never deploy / never do)

From `context/_global.md §Anti-patterns`. Each rejection has a documented rationale elsewhere; these exist to short-circuit any "but what if we used X" discussion.

- EC2 of any size (burns credits; PC is free).
- Amazon Managed Grafana (\$9/user/month; local Grafana matches).
- Kinesis Firehose (no Always-Free tier).
- Glue Crawlers (cost per run; schema-in-Terraform achieves the same).
- Timestream-for-InfluxDB (RDS-style billing; ~\$50/mo floor).
- Anything outside eu-central-1 during demo (cross-region transfer cost).

Where to look next

Workflow + roles

DEV_NORMS.md

Context-loading rules (Tier 1/2/2b)

DEV_NORMS.md §5

Component being worked on

context/<component>.md

Cross-component data shapes

context/_interfaces.md

Past decisions

docs/adr/ — especially ADR 0005 if your session touches the parity boundary

Past sessions

docs/sessions/

Open Qs from project planning

HANDOFF.md §6

Master plan

PLAN.md (currently .docx; consider docx->md cleanup)